

HCI and Design

Today

Design patterns and components

Material Design (Google)

Basically: How can YOU create designs that look good?

Design = Solutions

Design is about finding solutions

Unfortunately, designers often reinvent things

Hard to know *how* things were done before

Why things were done a certain way

How to reuse solutions

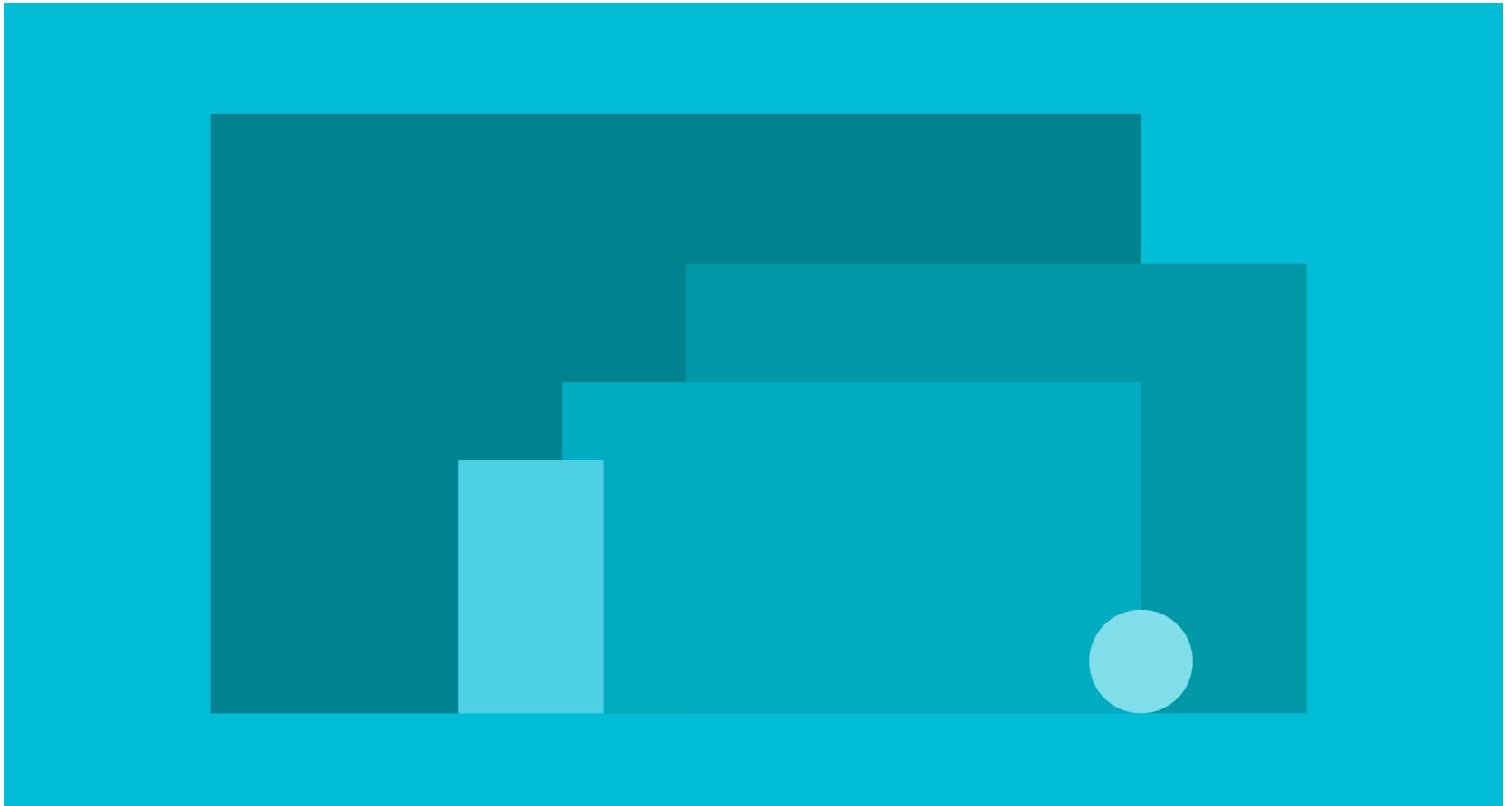
How can codify design knowledge?

- An effective and flexible design is difficult to get “right” the first time.
- Yet experienced designers do make good designs
 - New designers (like many of you!) are usually overwhelmed by the all the design options available.
- Experienced designers evidently know something inexperienced ones don't, what is it?

How can codify design knowledge?

- Expert designers usually do **not** solve every problem from first principles, they **reuse** solutions that have worked for them in the past.
- When they find a good solution, they use it again and again.
- How can you reuse solutions that good designers have used again and again?
 - But still make them new/yours
 - And learn how to create awesome designs at the same time

Answer: Design patterns



Web design circa 2006

Amazon.co.uk
MasterCard
Apply now
more info

amazon.co.uk

VIEW BASKET | WISH LIST | YOUR ACCOUNT | HELP

WELCOME | YOUR STORE | BOOKS | ELECTRONICS & PHOTO | MUSIC | DVD | VIDEO | SOFTWARE | PC & VIDEO GAMES | HOME & GARDEN | TOYS & GAMES

INTERNATIONAL | HOW TO ORDER | GIFT CERTIFICATES | SELL YOUR STUFF | HOTTEST OFFERS | HARRY POTTER

One for the ladies: get the pink [Motorola V3 mobile phone](#) on Virgin Pay As You Go and see more [pre-pay mobiles](#).

STAR CHOICE
Up to 50% Off Erotica
Treat yourself to some satisfying *Erotica*, all at up to 50% off, including [How to Be a Sex Goddess](#)
Our Price: £5.99

BROWSE

- Books
- DIY & Tools
- DVD
- DVD Rental
- Electronics & Photo
- Garden & Outdoors
- Gifts
- Home & Garden
- Kitchen & Home
- Music
- PC & Video Games
- Personal Care
- Software
- Toys & Games
- Video

- Used
- Auctions
- zShops

Add Favourites

virgin.net
Broadband

Broadband with no funny stuff.
From £14.99 a month.
No 12 month contract.
Free £10 Gift Certificate.

STEPHEN KING
45% off

Books

- 35% off Deals of the Week
- 3 for £ on 100s of paperbacks
- Up to 50% off New Releases
- Up to 50% off Business books

Electronics & Photo

- MP3 players
- Digital cameras
- DVD players and recorders
- Kick out your computer

PC & Video Games

- 25% off new releases
- £5 off in Special Offers
- Hardware and accessories
- Hot 100 games

CONSTANT GARDENER
40% off

DVD

- Up to 40% off Bestsellers
- 4 DVDs for £
- £5 off DVD Special Offers
- Hot 100 DVDs

Home & Garden

- More to Explore: Software
- More to Explore: Toys & Games
- Find all our hottest offers

SONY NWA Walkman
The new [Sony NWA Walkman](#) combines high-end design with ease-of-use and advanced functionality which seriously upgrades your listening experience.
Our Price: £130.49

Exclusive Deals
Check out our range of exclusive 3CD compilations, including [Essential 80s](#) by Various Artists.
Our Price: £8.99

WHAT'S NEW
Already a customer?
[Sign in](#) to see your **New for You™** recommendations.

Your Shopping Basket
You have 0 items in your Shopping Basket.

New Releases

Jan 2007: The iPhone is introduced



The crisis of a tiny screen

Everyone simultaneously realizes their website is completely unusable.



But Apple's own apps worked pretty well...



iOS DEV KIT

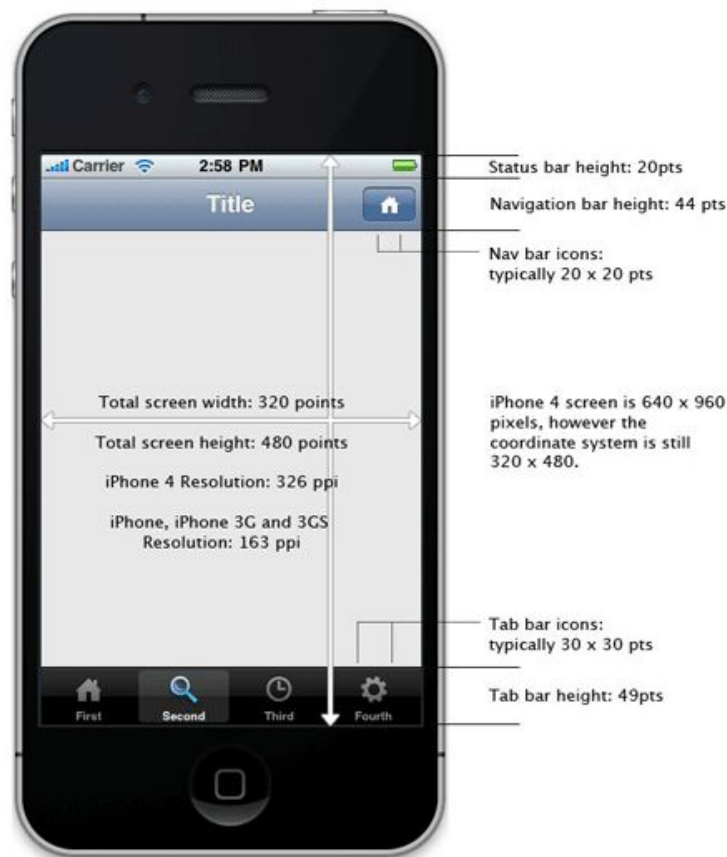
Guidelines for plug & play UI



Apple as the benevolent dictator

An opportunity to **redesign**
EVERYTHING from
scratch...

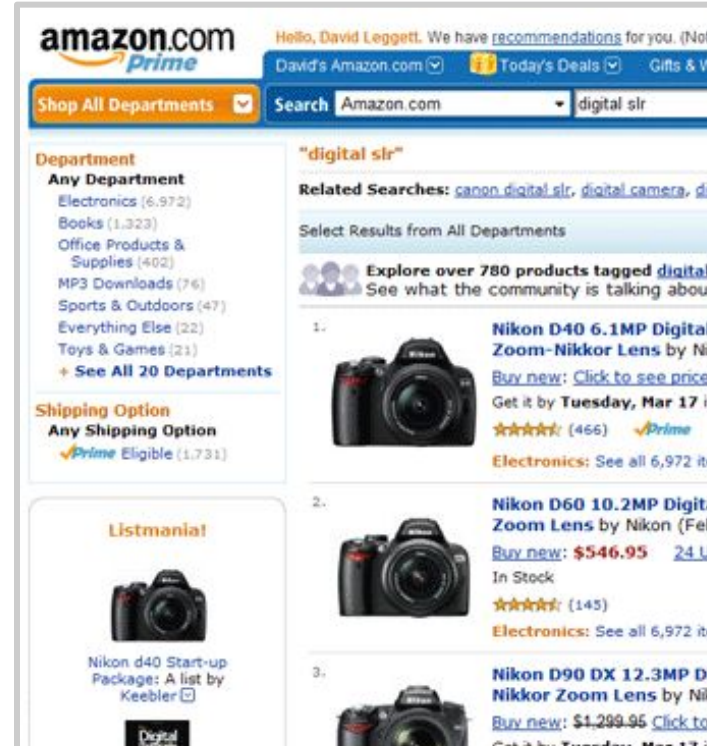
(for a very specific device)



The resource-rich fared best

Re-make your app 10 times!

- Web
- iPhone
- iPad
- Android
- Windows Phone
- Smart TVs
- etc...



A SOLUTION

Responsive design for any screen



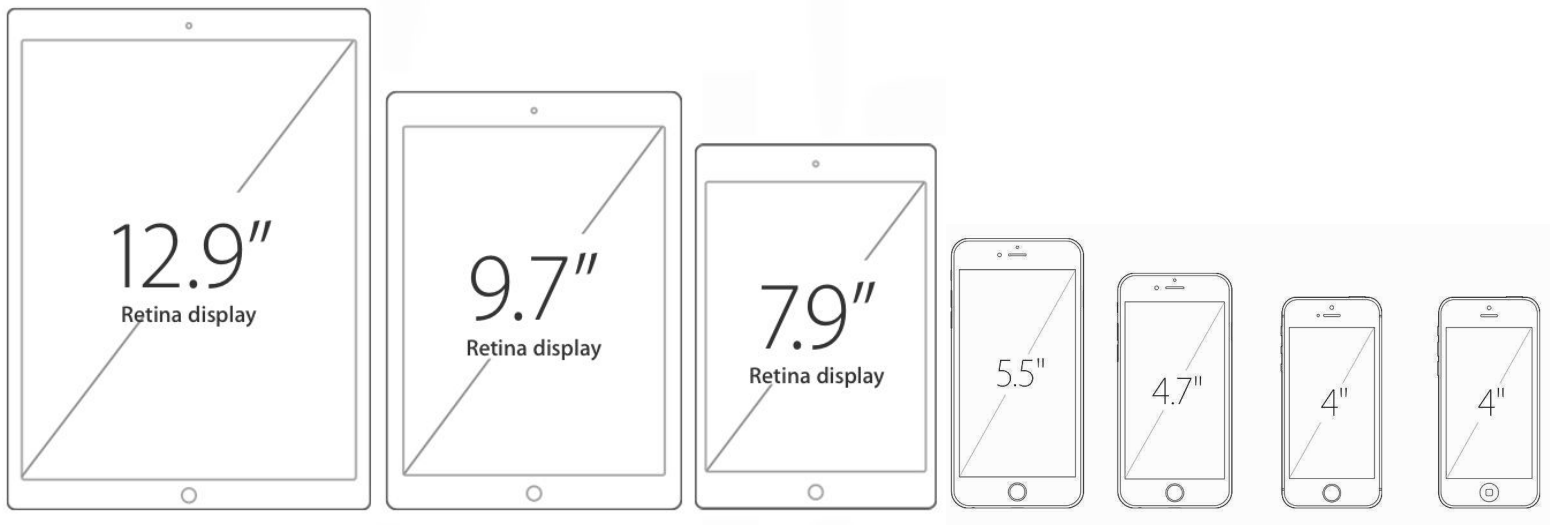
Adaptability also a mobile issue...

Oh Android. You so crazy.

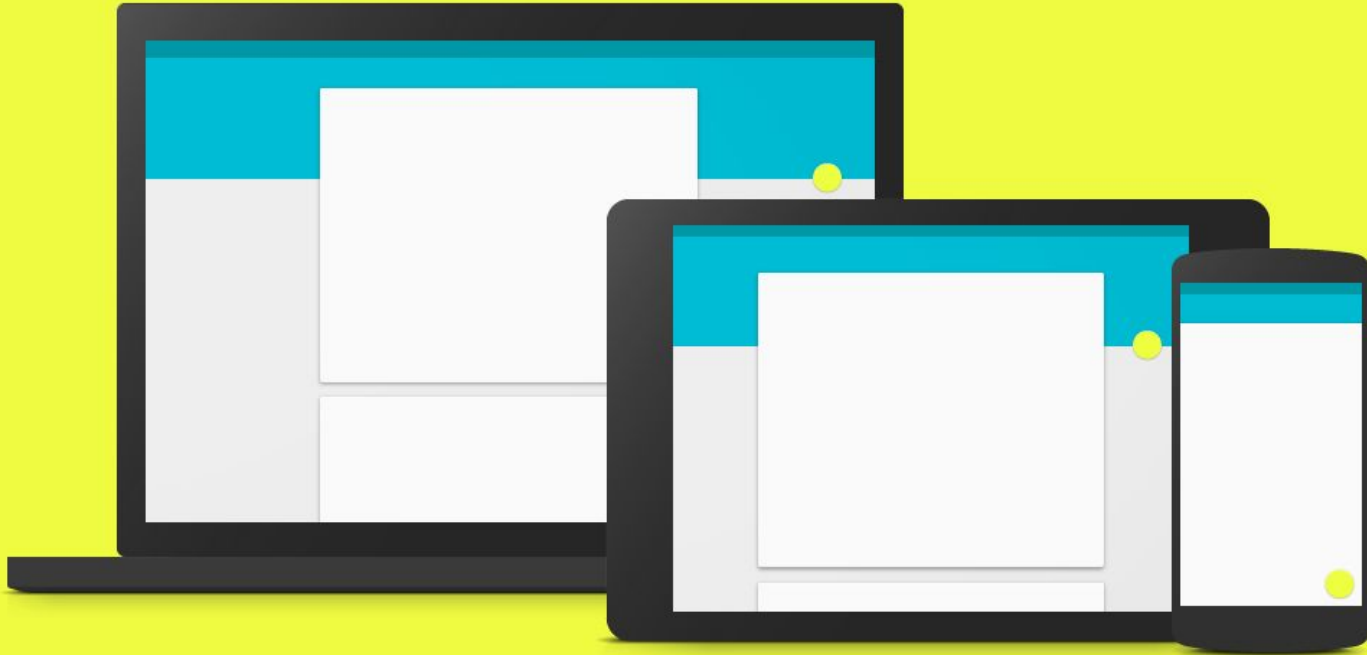


...eventually even for Apple

No more 320x480 pixel-perfect layouts :-)



Google responds: Material Design



Material Design

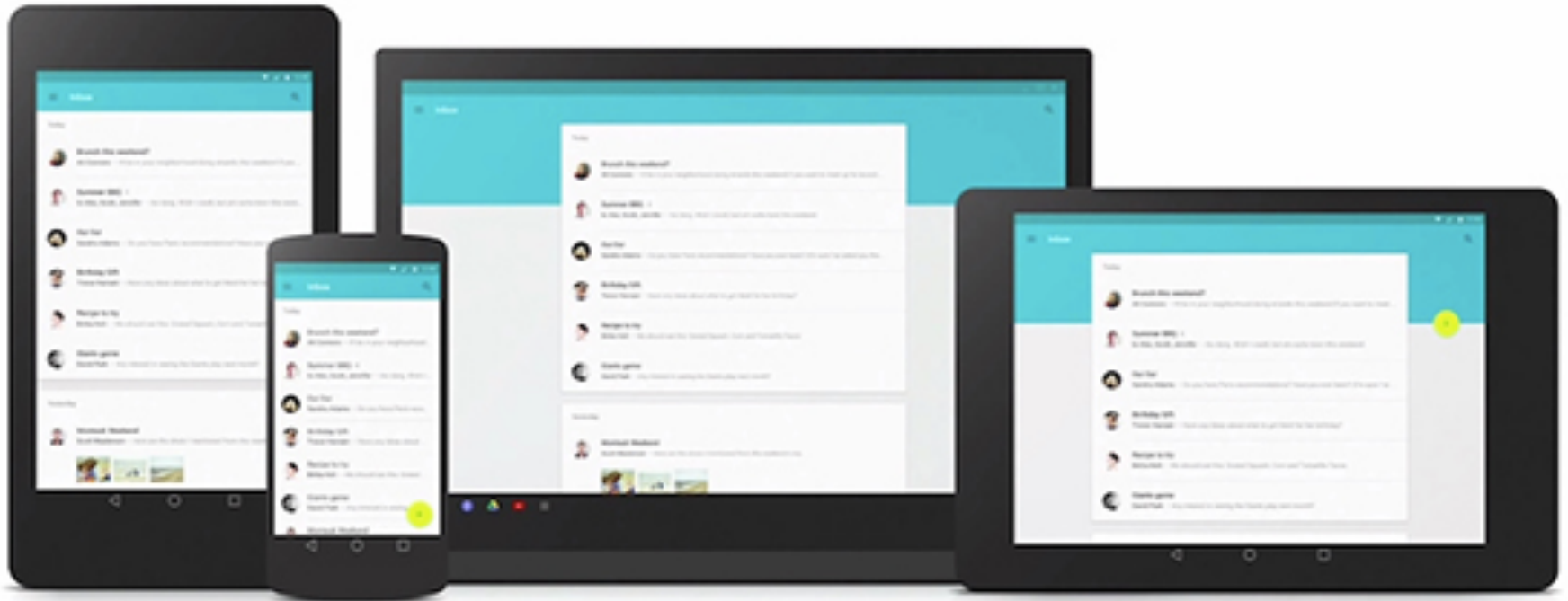
Released by Google in 2014

material.io

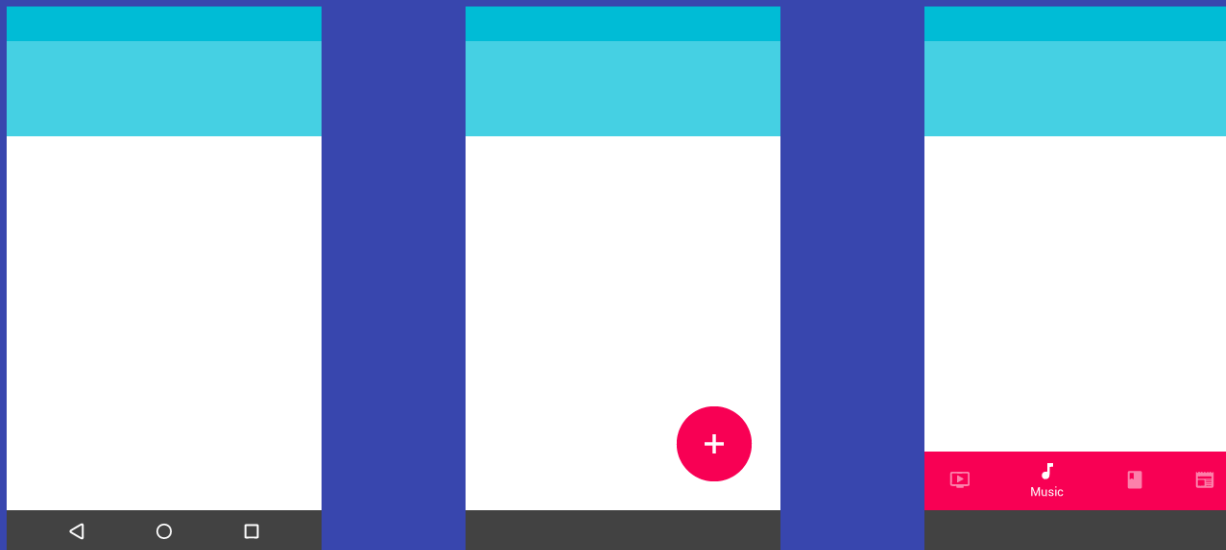
The main purpose of material design is to create a new visual language that combines principles of good design with technical and scientific innovation.

Develop a single underlying system that allows for a unified experience across platforms and device sizes.

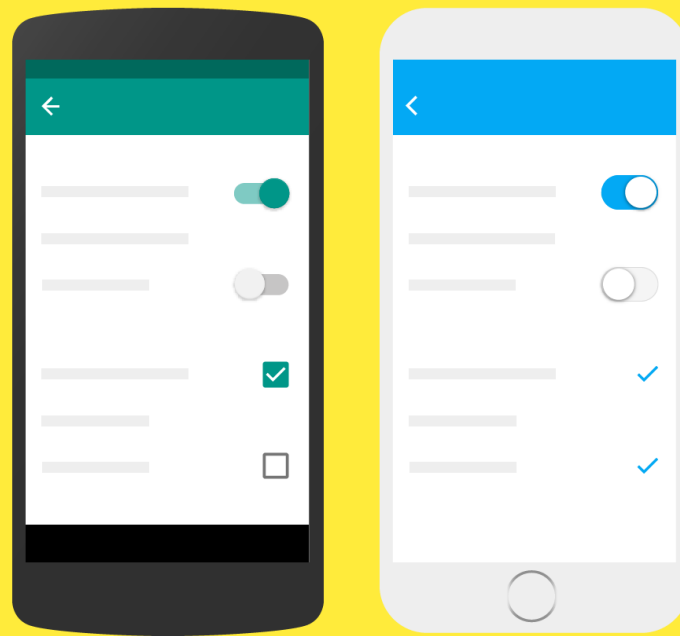
Material Design



Reusable components



Different devices



Buttons



Floating action button



Raised button

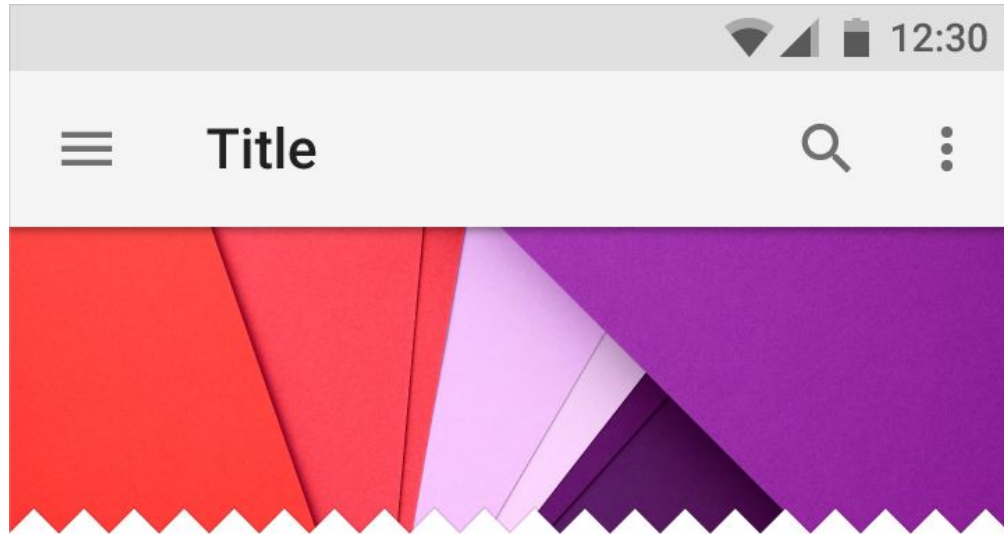
BUTTON

Flat button

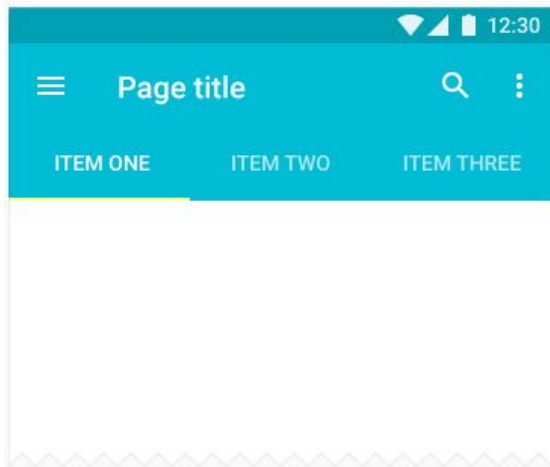


Icon buttons

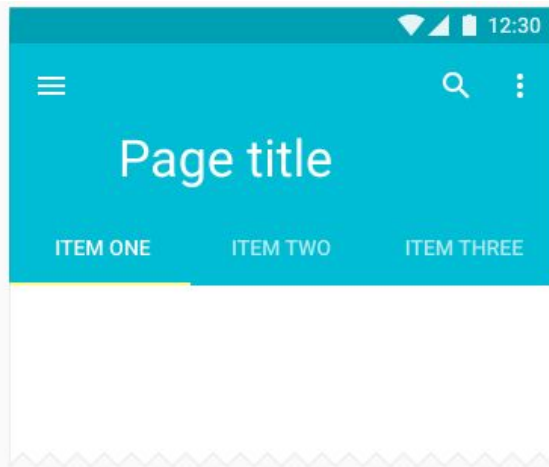
Navigation bar



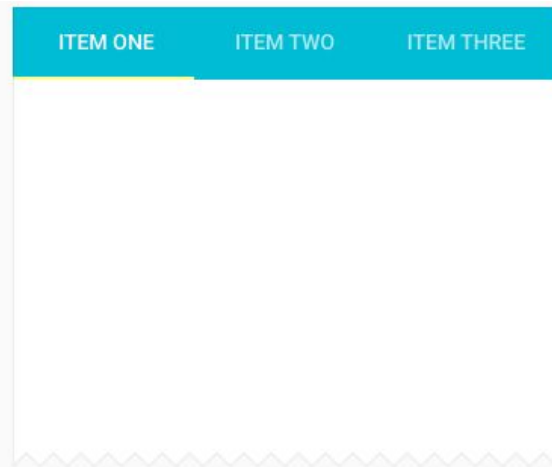
Tabs



Default app bar and fixed tab bar

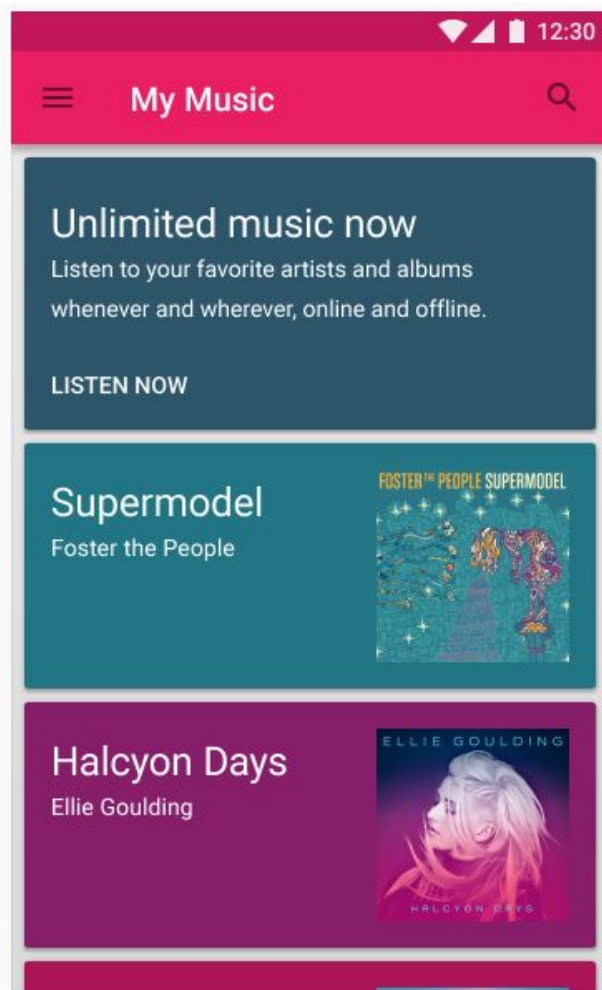
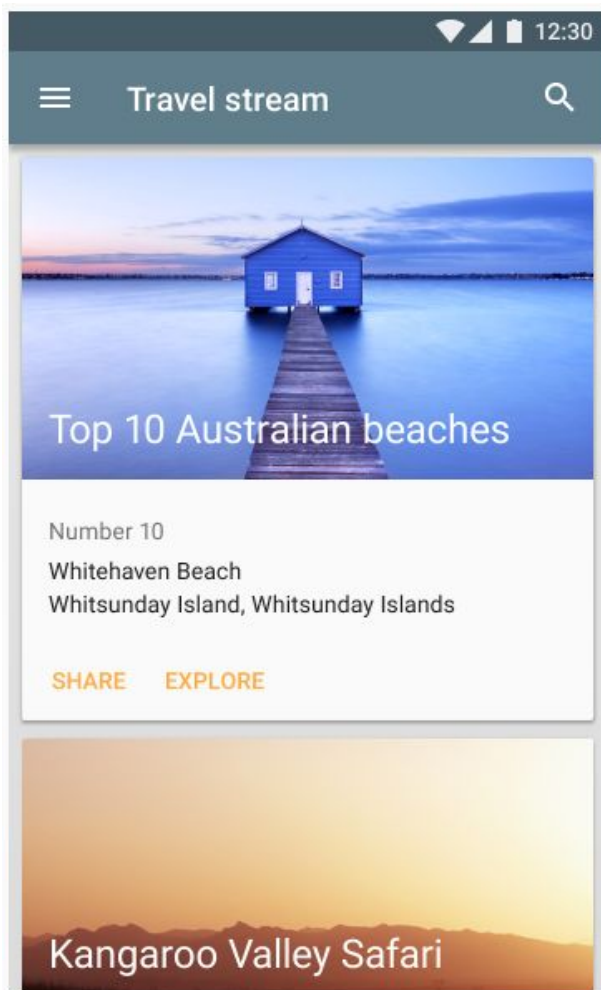


Extended app bar and fixed tab bar



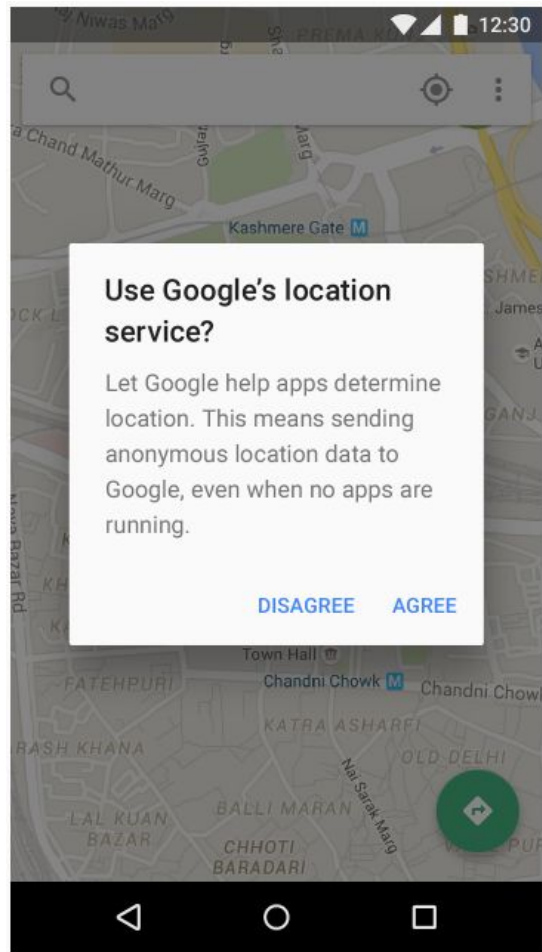
Fixed tab bar pinned to top with scrolled content

Cards



Dialogue

Spot the <card> and the
<button>!



**Any app
can be
reduced to
these basic
elements**

